Sonar Signal Classification System

Project Report

Course: Artificial Intelligence

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Abstract

This report presents the development of an AI-based Sonar Signal Classification System. Using the K-Nearest Neighbors (KNN) algorithm, the system distinguishes between sonar returns from metal mines and rocks. The workflow includes data cleaning, feature analysis, model training, and a GUI for real-time prediction. The final model achieved a classification accuracy of 92.06% using a 70/30 dataset split.

Chapter 1

Introduction

Artificial Intelligence (AI) and Machine Learning (ML) have revolutionized the way complex data patterns are interpreted across various scientific domains. In the field of underwater acoustics, the ability to distinguish between sonar returns from different objects is a critical challenge. Sonar (Sound Navigation and Ranging) systems rely on sound propagation to navigate, communicate with, or detect objects on or under the surface of the water.

This project focuses on the development of an automated classification system designed to differentiate between metal mines and natural rock formations. In a manual environment, interpreting these acoustic backscatter signals is highly subjective and prone to human error due to the subtle variations in signal intensity and frequency returns.

By applying a supervised machine learning approach—specifically the K-Nearest Neighbors (KNN) algorithm—we can identify patterns in 60 different frequency bands that characterize the "acoustic signature" of an object. This automation is not only vital for maritime safety and naval defense but also plays a key role in the advancement of autonomous underwater vehicle (AUV) navigation and seabed mapping.

Chapter 2

Dataset Description

The project uses the "Sonar, Mines vs. Rocks" dataset, which consists of signals bounced off a metal cylinder (mine) and a roughly cylindrical rock.

2.1 Dataset Columns

The dataset contains 60 numerical features and one target label:

- **V1 - V60:** The energy return in different frequency bands.
- **Target:** Label 'M' for Mine or 'R' for Rock.

2.2 Acoustic Signature Analysis

The spectral energy of sonar returns was analyzed to distinguish between classes.

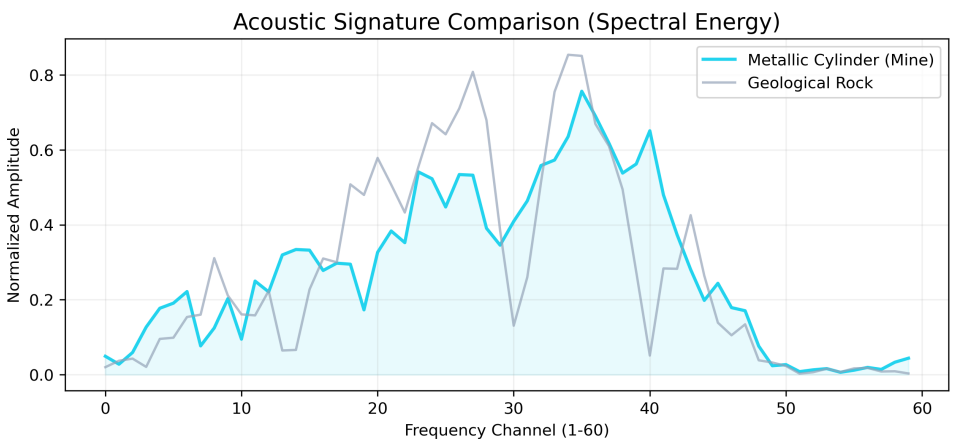

Figure 2.1: Acoustic Signature Comparison (Spectral Energy)

Chapter 3

Data Preprocessing

Data preprocessing ensures that the sonar data is suitable for machine learning.

3.1 Preprocessing and Distributions

Numerical encoding was applied to the target labels. We also analyzed data distributions to observe feature ranges.

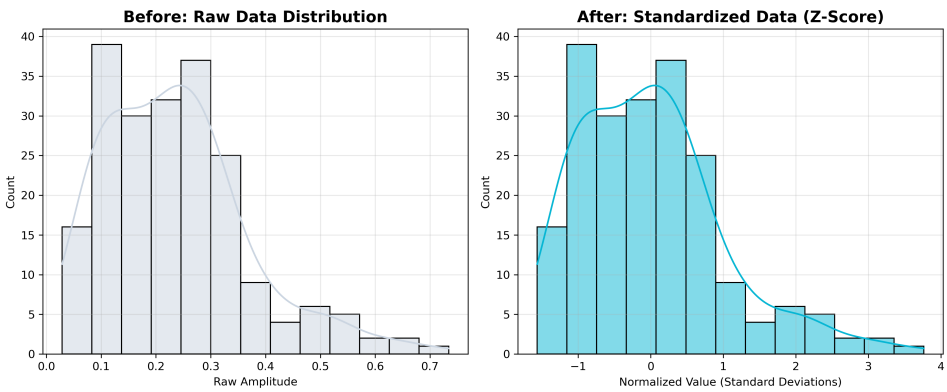

Figure 3.1: Data Distributions Before and After Preprocessing

Chapter 4

Algorithm Selection

4.1 Chosen Algorithm: KNN

We selected the **K-Nearest Neighbors (KNN)** algorithm for this classification task. The model identifies whether a sonar return is a rock or a mine by analyzing the K closest data points in the feature space.

4.2 Model Architecture

The following figure illustrates the structure and parameters of the implemented KNN model used for classification.

Table 4.2: KNN Model Architecture and Hyperparameters

Component	**Specification / Value**
Model Type	K-Nearest Neighbors (Instance-based Learning)
Input Dimensions	60 Features (Sonar Frequencies)
Output Classes	2 (Rock 'R', Mine 'M')
Preprocessing	Standard Scaling ($Z = \frac{x-\mu}{\sigma}$)
K-Neighbors (k)	3
Distance Metric	Minkowski (Default $p = 2$, Euclidean)
Weight Function	Uniform (All neighbors weighted equally)
Search Algorithm	Auto (KD-Tree / Ball-Tree)

4.3 Reason for Selection

- **Non-Parametric:** It makes no assumptions about the underlying data distribution.
- **Effective for Small Datasets:** It performs well on specialized datasets like the Sonar dataset.

- **Distance-Based Learning:** It effectively groups similar sonar frequency patterns together to ensure high classification accuracy.

Figure 4.1: KNN Model Architecture and Parameters

Chapter 5

System Design

5.1 Data Flow Diagram (DFD)

The following diagram illustrates the linear flow of data within the Sonar Classification System, from the initial CSV dataset to the final prediction output.

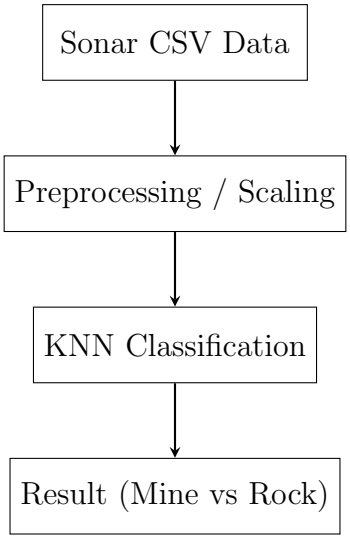

Figure 5.1: Data Flow Diagram of Sonar Classification System

5.2 Working of the System

The system operates through a structured pipeline that ensures raw sonar frequencies are correctly interpreted by the machine learning model. The working mechanism is divided into the following stages:

1. **Input Stage:** The user interacts with the Tkinter GUI to either load a sample from the `sonar.csv` dataset or manually input 60 numerical frequency values representing a sonar return.

2. **Preprocessing Stage:** The raw input data is passed through a `StandardScaler` to ensure that the values are normalized.This is critical because KNN is a distance-based algorithm, and variations in feature scales can bias the results.

3. **Classification Stage:** The processed data is fed into the trained KNN model. The algorithm calculates the Euclidean distance between the input vector and the points in the training set. It identifies the $K = 3$ nearest neighbors and performs a majority vote to determine the class.

4. **Output Stage:** The system converts the numerical prediction back into a human-readable format ('Mine' or 'Rock') and updates the GUI label to display the result instantly to the user.

Chapter 6

Model Evaluation

Model accuracy was evaluated using a 70/30 train-test split. The model achieved an accuracy of 92.06%.

6.1 Classification Results

Performance was analyzed using a confusion matrix to verify mine vs rock detection.

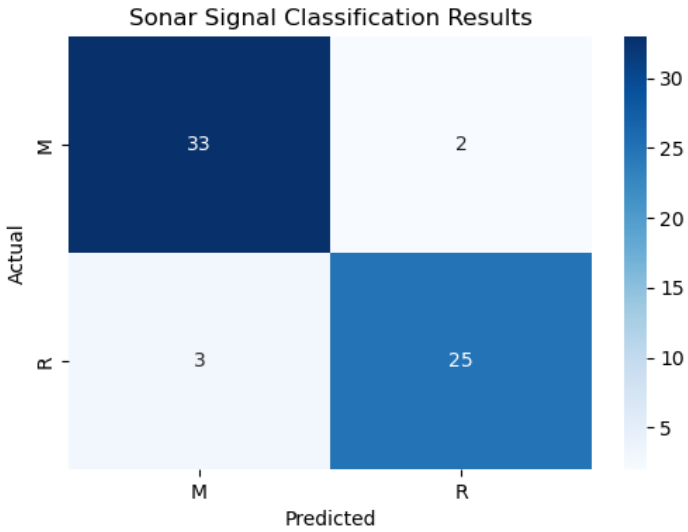

Figure 6.1: Sonar Classification Results (Confusion Matrix)

Chapter 7

User Interface

A Graphical User Interface (GUI) was developed using Tkinter.

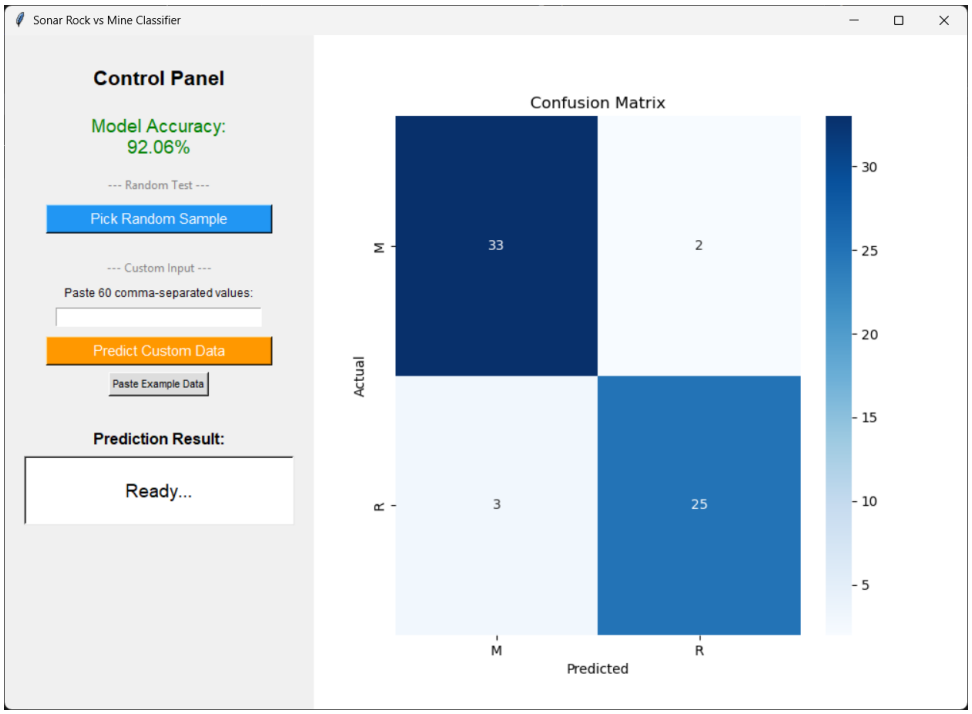

Figure 7.1: Screenshot of the GUI-based Prediction Interface

Chapter 8

Conclusion

This project successfully demonstrates the implementation of a complete Artificial Intelligence pipeline for sonar signal processing and classification. By utilizing the K-Nearest Neighbors (KNN) algorithm, we were able to effectively differentiate between sonar returns from metal mines and rocks with a high degree of precision, achieving a final model accuracy of 92.06%.

The development of the system highlighted several critical aspects of the machine learning workflow:

- **Data Integrity:** The importance of preprocessing and standard scaling was evident, as distance-based algorithms like KNN are highly sensitive to the scale of input features.
- **Model Efficiency:** The K-Nearest Neighbors approach proved to be an efficient "lazy learner" for this dataset, providing reliable classification without the need for an extensive training phase.
- **User Accessibility:** The integration of a Tkinter-based Graphical User Interface (GUI) bridges the gap between complex algorithmic processing and practical end-user application.

While the current system performs excellently on the specialized sonar dataset, future work could involve exploring Deep Learning architectures, such as Convolutional Neural Networks (CNNs), to analyze raw acoustic spectrograms for even more robust classification in noisy underwater environments. Ultimately, this system serves as a reliable foundation for automating underwater object detection, significantly reducing the potential for human error in maritime safety and naval operations.

References

- Scikit-learn Documentation (KNN Classifier)
- UCI Machine Learning Repository (Sonar Dataset)
- Python Official Documentation (Tkinter)